

2 MISCELLANEOUS CONTROL

2.1 Developmental Model Control, 1

This card has been added to the code for the convenience of developers in testing model improvements or new models. This card is not a standard input feature of the code. The description of this card has been added to the input requirements because several laboratories are receiving test versions to assist in the development and testing of the code. Anyone using this card must realize that they are selecting experimental options still under development. Furthermore, these options may change more frequently than the revision of this input manual. Thus, before using the options, users should obtain the brief listing of current options from the code (described below) and verify those descriptions against this manual.

The purpose of this card is to allow developers and analysts to quickly test new models by activating or deactivating a model through simple input instead of program modification, compilation, and loading. Ninety logical variables having only false or true values are provided and defined at the start of program execution as false. This input sets the logical variables to true or resets them to false at the beginning of a new problem or at any restart. Fortran IF statements added as part of the experimental coding activate or deactivate models based on the values of the logical variables.

As described above, up to 90 options can be defined and the options are identified with a number from 1 through 90. Which options are defined and what they control are very much version dependent. The usual practice is to enter the option capability using a currently unused option number as the new model or improvement is first coded. During further development and testing, the model may change and the effect of the option can change in a manner ranging from large to subtle. When the model has been completed or even abandoned, the production version of the model is coded and the option capability is removed. The option number is then available for reuse with a completely different model. Thus, the options are version dependent as to what option numbers are in use, what models they control, and the particular features of the models. Accordingly, these options should be used only by those in direct contact with the developers.

Each current option is described below. In light of the discussion above, the user should verify that the code version being used corresponds to this description. Programmers using this option feature are asked to include coding that issues error messages when unused options are selected and to issue a brief statement of the purpose of selected options. Remember, however, that all coding associated with these options is experimental and these output conventions may not be thoroughly checked.

Up to 91 numbers consisting of 0 or any of the currently available option numbers may be entered on this card. A positive nonzero number, n , activates option n by setting the logical variable n to true; a negative nonzero number, $-n$, deactivates option n by setting the logical variable n to false. Attempting to activate an unused option is an error, and attempting to deactivate an already inactive option or an unused option is also an error. Adding options or deactivating options is allowed at a restart; the previously defined options will remain. The status of the options is printed in any NEW or RESTART problem containing this card or a RESTART problem in which the restart point had an option selected. The printout

includes a listing of the 90 option numbers and a false (option not selected) or a true (option selected) value plus the brief description of each selected option.

The number 0 is not an option number but may be entered to force the brief descriptions of all available options to be printed regardless of whether they are active. The 0 input should be used only once to observe the available options and then removed so that the list better emphasizes the selected options.

W1-91. Option number as described above.

- | | |
|-----------|--|
| Option 0. | This option lists out all the available options. |
| Option 1. | This option adds an explicit viscous-like term to the momentum equations in the nearlyimplicit scheme, This term is documented in the discussion of the semi-implicit scheme. |
| Option 2. | This option sets the interfacial mass transfer to 0 and the direct heating heat transfer coefficient to 105. This option simulates the no mass transfer case in which the two phases are in thermal equilibrium. |
| Option 4. | This option scales back the Courant limit to 75% of its calculated value so that the time step will always be less than or equal to 75% of the Courant limit. |
| Option 6. | This option outputs information to the screen and to the screen file every time dtstep is called. It is useful for determining why the code is cutting the time step. The volume number for the volume that has the limiting Courant time step and the maximum mass error for each time step is printed on one line. In addition, the pressure and void fraction for each of these volumes are printed. If the time step is cut, the reason for the cut is given in the right-hand column. |
| Option 8. | This option provides time step control based on the change in void fraction and is designed to limit the time step when the void fraction in any cell is decreasing rapidly such as during periods of condensation. The time step will be repeated if the decrease in void fraction in any hydrodynamic volume is considered too large. The time step will be repeated . |
| Option 9. | This option provides a transition to plug flow when the vapor velocity exceeds the criterion for transition from stratified flow in horizontal geometries. The need for this change occurs during the reflood phase in the horizontal core of the Hanford N-Reactor. The process tubes in the core region are long and have a small diameter. As subcooled coolant enters the process tubes, high condensation occurs that results in local depressurization and high steam velocities towards the condensation site. Instabilities occur that eventually cause the code to fail. This changes introduces a plug flow model that limits condensation to a value that is just large enough to condense all the vapor that |

can flow at the critical velocity, utilizing the full channel cross-sectional area. The critical velocity is the velocity large enough to cause transition from a stratified flow condition. The concept behind this model is that condensation lowers the local pressure and draws in steam which pushes the water into a plug. The area of the plug limits condensation and reduces steam flow which causes a return to stratified flow.

- Option 10. This option provides time step control based on change in pressure within a hydrodynamic volume. This change causes the code to repeat a time step if the change in pressure during a time step exceeds the old time value, the new time value, or 50,000 Pa. This time step control allows the pressure to change by no more than a factor of two during a time step. This change generally causes the code to run slower but more reliably. With this change activated, the code will more accurately track pressure waves and oscillations but may cause certain problems to run unacceptably slowly. This option has no effect if Option 8 is not selected.
- Option 12. This option provides a user controlled (on/off) water packer developed for horizontal reactors such as the Hanford N-Reactor. The interfacial friction coefficient term for the momentum equation, C_i , is adjusted as a function of vapor void fraction. For void fractions of 0.001 or less, C_i is forced to an arbitrarily large value of 1010. For void fractions greater than 0.01, the regularly calculated C_i is used. For void fractions between 0.001 and 0.01, a cubic interpolation scheme is used to adjust C_i between the calculated value and 1010. As a cell is calculated to fill with liquid and the calculated cell pressure rises, the lower inertia vapor is the phase first to respond, either moving on to the next cell or moving back to an upstream cell. The effect of this model is that, as the void fraction decreases, the interfacial drag is increased, thus allowing the moving vapor to either drag liquid on to the next cell or impede the incoming liquid from an upstream cell. In either case, the model eases the overfilling and overpressurization of the cell.
- Option 13. This option activates vertical stratification changes.
- Option 14. This option turns off constitutive relations and should only be used for testing advancement of the basic advancement scheme for two-phase conditions. Do not use for single phase conditions.
- Option 16. This option activates the velocity flip/flop logic for the nearly-implicit advancement scheme.
- Option 17. This option allows the code to run at the Courant limit with a multiplication factor of four instead of two and a reduction factor of 0.25 instead of 0.5 used in the mass error check for the time step control.
- Option 18. This option adds the sharp interface and reverse void profile logic from MOD2.5.

- Option 19. This option uses the Bestion correlation (in MOD2.5) rather than the EPRI correlation (in MOD3) for bundles (when volume flag $b = 1$).
- Option 20. This option uses the annulus multiplication factor for bubble rise velocity. This model change was raised due to downcomer boiling issue for LBLOCA of DVI plant.
- Option 21. Do not use. Presently option only acquires additional storage arrays for testing numerical techniques.
- Option 23. This option selects a boron transport algorithm (Godunov) that greatly reduces the numerical diffusion of boron compared to the standard algorithm.
- Option 24. This option activates the original RELAP5 subcooled boiling model. The default subcooled boiling model is the SRL model, which is better at low pressures.
- Option 25. This option linearizes the interfacial heat transfer for the nearly implicit scheme.
- Option 32. This option activates the stretch logic for water packing model.
- Option 36. Option 36 limits metastable extrapolation to 50 K.
- Option 37. This option turns off the umbrella model. When the umbrella model is on, an upper limit is placed on the liquid interfacial heat transfer coefficient (H_{if}) when the liquid is subcooled. The limit is umbrella shaped so as to force the coefficient to small values as the void fraction approaches 0.0 or 1.0.
- Option 41. This option includes energy dissipation due to form loss.
- Option 42. This option applies a stronger unchoking test for junctions with an abrupt area change.
- Option 43. This option uses iteration to calculate the hydrodynamic conditions at the throat for critical flow of vapor.
- Option 45. This option selects the newly developed model for condensing interphase heat transfer.
- Option 48. This option requests use of the level tracking model during moving problems.
- Option 50. This option activates at all junctions where choking option is activated the original RELAP5 critical flow model instead of the default Henry-Fauske critical flow model. The code does not allow a user to switch between the two critical flow models in the middle of a problem, e.g. switching from the Henry-Fauske to the original critical flow model after a steady-state run at the beginning of a transient run.

- Option 51. Normally, water packing is activated in all volumes unless specifically disabled by an input volume flag. This option disables water packing for all volumes.
- Option 52. Normally, the choking model is activated for all junctions unless specifically disabled by an input junction flag. This option disables the choking model for all junctions.
- Option 53. This option activates the Moody critical flow model. It is part of the CANDU package.
- Option 54. This option activates the M.S.Chung modified mechanical critical flow model.
- Option 55. This option activates the HF critical flow based on Enthalpy Upstream.
- Option 56. This option enforces $v_g = v_f$ at $g = 1$ in BubbleDropDrag(FIDIS2).
- Option 57. Turbulent viscous stress terms in 3D channel and gap module is activated.
- Option 61. This option activates the Podowski subcooled boiling model instead of the Lahey method.
- Option 62. This option uses newly developed changes to the Chen F factor in PREDNB to minimize oscillations.
- Option 64. This option uses the old halving and doubling of the time step but the new Courant limit.
- Option 65. This option changes the subcooled boiling model by modifying the fraction of nucleate boiling heat flux that generates vapor when the bulk liquid is subcooled. The modification minimizes the “on/off” behavior associated with low-pressure/low-flow conditions.
- Option 66. This option implements donor/acceptor differencing in vertical stratification volumes.
- Option 67. This option activate the lateral momentum flux only for MULTID component. Using this option, the diffusion term of MULTID component is neglected.
- Option 68. This option use serial X-Y volume-junction ID number for MULTID Component
- Option 69. This option causes Two Step Method using IPF Model. Virtual Mass Force is deactivated
- Option 70. This option use ASME 67 type Dynamic property instead of ASME 93 property.
- Option 71. This option turns on the TMDPVOL momentum flux term.
- Option 75. This option causes the vaporization smoothing.
- Option 76. This option causes the default junction diameter as a min. volume hydraulic diameter.

- Option 77. This option activates the SNU bubble regime model in annulus.
- Option 78. This option deactivates 3D water-packing scheme.
- Option 80. This option enables the moving reactor model by KAERI
- Option 81. This option activates the dryout model using critical thickness correlation.
- Option 82. This option activates 3D dial.
- Option 83. This option activates generation of MARS data file for interfacing with MIDAS severe accident code.
- Option 84. This option activates implicit gravity head and horizontal stratification force due to the void gradient in the nearly implicit scheme.
- Option 85. This option produce tecplt.dat file of MULTID output parameter for TECPLOT 3rd party graphic software.
- Option 86. This option deactivates option 60. It restores the old Courant time step calculation using void weighted volume averaged velocities with icoran = 1.
- Option 87. This option deactivates the NC gas equation in the 3D module and a minimum internal energy of gas phase in the 1D module is set to 2.3e6 J/kg.
- Option 88. uses K KRB1 CHF correlation at P>50 bar rather than AECL Lookup Table
- Option 89. uses KNOEBEL Annuli CHF correlation in annuli geometry (120)
- Option 90. This option turns on the e-flag (modified PV term in the energy equation) for all applicable junctions. This includes all junctions except for time-dependent junctions, accumulator junctions, and junctions associated with separator, jetmixer, pump, turbine, and eccmixer components.
- Option 99. This turns on Iterative Matrix solver for sparse matrix.

2.2 Multiplier for the Global Models, 10

This input card was added for implementing the multipliers for the two-phase wall friction and the interfacial drag models, Ishii-Mishima entrainment coefficient, and We number of reflood, which can be used for the uncertainty quantification of MARS code.

Up to 10 multipliers can be entered in Card 10 and, at present, only two multipliers are used. If Card 10 is not entered, the default multipliers 1.0 are used, which means no changes in the two-phase pressure drop and interfacial drag models.

W1(R)	Multiplier for the two-phase wall friction.
W2(R)	Multiplier for the interfacial drag model.
W3(R)	Multiplier for the Ishii-Mishima entrainment coefficient.
W4(R)	Multiplier for We number of reflood.

2.3 Multiplier for the Component-wise two phase wall friction, 11

This input card was added for implementing the multipliers for the two-phase wall friction for the specified components. Up to 9 components can be entered in card. If card is not entered, the default multipliers 1.0 are used.

W1(R)	Multiplier.
W2(I)	Component 1 identification number: CCC
W3(R)	Component 2 identification number: CCC
W4(R)	Component 3 identification number: CCC
...	
WN(I)	Component N identification number: CCC

2.4 Multiplier for the component-wise interfacial drag model, 12

This input card was added for implementing the multipliers for the interfacial drag model for the specified components. Up to 9 components can be entered in this card. If card is not entered, the default multipliers 1.0 are used.

W1(R)	multiplier.
W2(I)	Component 1 identification number: CCC
W3(R)	Component 2 identification number: CCC
W4(R)	Component 3 identification number: CCC

...

WN(I) Component N identification number: CCC

2.5 Multiplier for the component-wise interfacial heat transfer model, 13

This input card was added for implementing the multipliers for the interfacial heat transfer model for the specified components. Up to 9 components can be entered in this card. If card is not entered, the default multipliers 1.0 are used.

W1(R) multiplier.

W2(I) Component 1 identification number: CCC

W3(R) Component 2 identification number: CCC

W4(R) Component 3 identification number: CCC

...

WN(I) Component N identification number: CCC

2.6 Problem Type and Option, 100

This card is always required.

W1(A) Problem type. Enter one of the following: NEW, RESTART, RESET, REEDIT, STRIP, or CMPCOMS.

NEW specifies a new simulation problem.

RESTART specifies continuation from some point in a previous problem using information from the RSTPLT file.

STRIP specifies that data are to be extracted (stripped) from the RSTPLT file, and only the data specified are written to the STRIP file.

W2(A) Problem option. This word is needed if W1 is NEW or RESTART and is optional if W1 is STRIP.

If NEW or RESTART is entered, in W1, enter either STDY-ST or TRANSNT in W2 to specify the type of simulation. Note the cautions discussed in Section 2.9 when the problem option is changed from STDY-ST to TRANSNT or vice versa.

When STRIP is entered in W1, enter either BINARY or FMTOU or leave W2 blank. BINARY is assumed if W2 is not entered. BINARY indicates the unformatted (BUFFER OUT) file. FMTOU indicates that the same information is to be written as 80-column formatted records. One use of this option is to allow simulation results to be transmitted to a different type of computer. Formats are

STRIP Record 1. (5A8,10X,A8)

STRIP Record 2. (A10,3I10)

STRIP Record 3. (8A10)

STRIP Record 4. (A10,7I10/(8I10))

STRIP Record 5..., N. (A10, 5X,1P,4E15.6/(5E15.6)).

The STRIP record above refers to the data in one record of the unformatted file. Multiple 80-column formatted records may be written for STRIP Records 3 through N.

2.7 Input Check or Run Option, 101

This card is optional for all types.

W1(A) Option. Enter either INP-CHK or RUN; if this card is omitted, RUN is assumed. This card has no effect on a CMPCOMS problem

If INP-CHK is entered, the problem execution stops at the end of input processing.

If RUN is entered, the problem is executed if no input errors are detected.

2.8 Units Selection, 102

This card is optional for all problem types. If the card is omitted, SI units are assumed for both input and output. If the card is used, enter either SI or BRITISH for each word. SI units used are the basic units, kg, m, s, and the basic combined units such as $\text{Pa} = \text{kg} \cdot \text{m} / \text{s}^2 \cdot \text{m}^2$. British units are a mixture of lb (mass), ft, and s, primarily, but pressure is in $\text{lb}_f / \text{in}^2$ (lb_f is pounds force), heat energy is in Btu, and power is in MW. Thermal conductivity and heat transfer units use s, not h.

W1(A) Input units.

W2(A) Output units. If this word is missing, SI units are assumed for output.

2.9 Restart Input File Control, 103

This card is required for all problem types (W1 of card 100) except NEW and is not allowed for type NEW.

When the problem option (W2 on card 100) is the same as the problem being restarted, the steady-state or transient is continued, and data on the RSTPLT file up to the point of restart are saved. If the restart continues from the point the previous problem terminated, restart and plot information is added to the end of the previous RSTPLT file. If the restart is prior to the termination point of the previous simulation, restart and plot data after the point of restart are overwritten by new results. A copy should be saved if RSTPLT files from each simulation are needed. If the problem options are different, data up to the point of restart are not saved, problem advancement time is reset to 0, and the RSTPLT file will contain information as if this problem type were NEW.

Some cautions should be observed when the problem advancement time is changed by changing the problem option from STDY-ST to TRANSNT, or vice versa, or the problem advancement time is reset through W1 on card 200. Either or both of these could be specified at restart. When the advancement time is changed, the user is responsible for ensuring that models involving problem time will operate as intended. Affected models include trips using advancement time, control systems using time as an operand (does not include differentiation or integration with respect to time), and table lookup and interpolation using time as the independent variable. If necessary, trips, control systems, general tables, time-dependent volumes, junctions, and pump speed tables can all be reentered at restart. With normal modeling practices, little use of modeling features involving advancement time is needed for runs to steady-state and accordingly little effort should be needed in switching from STDY-ST to TRANSNT. Because of the frequent use of time in logic to initiate failures, as part of safety systems, and used in establishing the delay times allowed in most table lookup and interpolation tables, required changes to a transient run may be extensive.

The program does make a change to delay control components when the advancement time is changed. The delay control component operates by maintaining a tabular past history of the delayed functions and using table lookup and interpolating to evaluate the delayed function. The table consists of pairs of time values and the delayed function. When the problem time is changed, the time values in the history table and the time value to store the next point in the table are modified by adding the difference of the new advancement time and the old advancement time. The modified history table is as if the problem being restarted was run with the new advancement time. This may not be the desired change, and, in that case, the user can reenter the delay component.

W1(I) Restart number. This must be a number printed in one of the restart print messages and whose associated restart information is stored in the RSTPLT file. If the problem type (W1 on card 100) is STRIP, this number must be 0.

Alternately, a -1 may be specified instead of a specific restart number. This has the effect of restarting a problem from the last block in the restart file. Thus, the user can now set up a series of restart cases which begin from the end of a previous run without having to wait for the run to finish in order to get the actual restart number from which to restart.

- W2-6(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. (The code internally treats the field as up to five eight-character words.) The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field.

2.10 Restart-Plot File Control, 104

This card can be entered for NEW, RESTART, and STRIP options. For the strip option, this card controls the strip file, and the NONE option is not allowed. Only the FILENAME option can be used. If this card is omitted, the restart-plot file is rewound at the end of the problem, but no further action is taken. The user may need to provide system control cards to dispose of the file. To prevent the restart-plot file from being written, a card with NONE must be entered.

- W1(A) Action. This word may not be blank. If this word is NONE, no restart-plot file is written. If this word is FILENAME, words 2-6 on this card contain the restart plot file name.
- W2-6(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. (The code internally treats the field as up to five eight-character words.) The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field. This information can be entered only on NEW problems; in RESTART problems, this information may be entered on the 103 card.

2.11 CPU Time Remaining and Diagnostic Edit, 105

Card 105 controls termination of the transient advancement based on the CPU time remaining for the job. Some operating systems allow specification of the CPU time allocated for a job as part of the job control language and also provide a means to determine the CPU time remaining during job execution. As an alternative, Word 3 of this card may be entered as the CPU time allocated. An alternative CPU remaining time is computed by decrementing this quantity by the CPU used as measured by the program. If Word 3 is omitted or 0, the alternative CPU remaining time is assumed infinite. At the end of each time step, the CPU time remaining for the job is determined from the minimum of the system (if available) and alternative CPU remaining times. If the remaining CPU time is less than Word 1, the transient is immediately terminated. The advancement may not be at the end of a requested time step due to time step

reduction; the hydrodynamic, heat conduction, and reactor kinetics may not be advanced to the same point; or the advancement may not be successful and the advancement is scheduled to be repeated with reduced time step. Major edits, minor edits, plot edits, and a restart record are forced. The transient can be restarted from this point as if the problem had not been interrupted. The transient is also terminated after successful advancement over a requested time step and the CPU time is less than Word 2. Word 2 should be larger than Word 1. The default values for Words 1 and 2 are 1.0 and 2.0 seconds. The default values are used if the card is not supplied or the entered numbers are less than default values. Word 2 is also forced to be 1.0 seconds larger than Word 1. The time values must include time for the final minor and major edits (very little time required), plotting, and any other processing that is to follow termination of RELAP5 execution. This card is optional, but we strongly recommend its use with Word 3 nonzero on systems that do not provide a system CPU limit.

Card 105 also controls the diagnostic edit printout through the use of Words 4 and 5. If these words are missing or 0, no debug options are in effect. If Word 4 is greater than 0, then Word 4 is the attempted advancement count number to start a diagnostic edit, and Word 5 is the attempted advancement count number to stop the diagnostic edit as well as the calculation. If Word 4 is -1, a dump file is written on the file specified by the -A option on the command line at the completion of the advancement given in Word 5. Entering 0 in Word 5 writes the dump file just before the start of transient advancement. The problem is terminated after writing the dump file. If Word 4 is -2, a dump file is written on the file given by the -A option after the advancement given in Word 5; the time advancement is then repeated and a dump file following the repeated advancement is written on the file given by the -B option. The problem is terminated after writing the second dump file. Word 5 must be greater than 0 when Word 4 is -2. The default file names are -A dumpfil1 and -B dumpfil2.

W1(R)	CPU remaining limit 1 (s).
W2(R)	CPU remaining limit 2 (s).
W3(R)	CPU time allocated (s). This quantity is optional.
W4(I)	Debug control word as described above.
W5(I)	Debug control word as described above.

If the program is compiled with compile time option CTSS defined, entering Word 1 as 0.0 will cause no testing for CPU termination and normal CTSS termination at the end of CPU time can occur. In this case, the problem can be restarted.

2.12 Noncondensable Gas Species, 110

This card is required for all calculations that use noncondensable gas. Nitrogen must be included for any problem having accumulators. This card cannot be entered on a RESTART problem.

W1-WN(A) Noncondensable gas type. Enter any number N of words (maximum 5) of the following noncondensable gas types: ARGON, HELIUM, HYDROGEN, NITROGEN, XENON, KRYPTON, AIR or SF6.

2.13 Noncondensable Mass Fractions

Card 115 is related to card 110. card 115 is required if card 110 is entered unless only one species is entered on card 110, and then the mass fraction is set to 1.0. The number of words on card 115 must equal the number of words on card 110. The sum of the mass fractions must sum to 1.0. The mass fractions on these cards are default values and are used for initial conditions of active volumes and for values of time-dependent volumes. This card cannot be entered on a RESTART problem.

2.13.1 Initial Mass Fraction for Each Noncondensable Gas Type, 115

W1-WN(R) Mass fraction for each noncondensable gas type.

2.14 Card 119, Gravity Constant

This card is optional and specifies the gravitational constant. If not entered, the earth gravitational constant of 9.80665 m/s^2 is used.

W1(R) Gravitational constant (m/s^2 , ft/s^2). A positive number, which must be greater than or equal to $1.010 \times 10^{-6} \text{ m/s}^2$ (or $3.2808 \times 10^{-6} \text{ ft/s}^2$ if British input is used), is used as the gravitational constant. If -1.0 is entered, the earth gravitational constant is used.

2.15 Hydrodynamic System Control

Independent hydrodynamic systems can be described by the hydrodynamic component input. The term independent hydrodynamic systems means that there is no possibility of flow between the independent systems. A typical example would be the primary and secondary systems in a reactor where heat flows from the primary system to the secondary system in the steam generator but there is no fluid connection. If a tube rupture were modeled, the two systems would no longer be independent. Input processing lists an elevation for each volume in each independent hydrodynamic system and includes a check on elevation closure for each loop within a system. A reference volume is established for each system through input or default.

These cards are optional for each system. If not entered for a system, that system contains H_2O as the fluid unless a different fluid is specified in hydrodynamic component data, and the lowest numbered volume in each system is the reference volume. Additionally, the reference volume has a default elevation of 0. These cards should not be entered in a RESTART problem.

2.15.1 Hydrodynamic System, 120-129

- W1(I) Reference volume number of the system. This must be a volume in the hydrodynamic system.
- W2(R) Reference elevation of the volume center relative to a fixed z-axis for the system (m, ft).
- W3(A) Fluid type for the system. Enter H2O or H2ONEW or D2O or HE for helium or CO2 for carbon dioxide. If H2ONEW is entered, the code uses the newer steam tables and if H2O is entered, the code uses the older thermodynamically-consistent tables. The newer tables are based on the Helmholtz functional fit in the IAPWS-1995 document.
- W4(A) Optional alphanumeric name of system used in output editing. *NONE* is used if this word not entered.
- W5(I) System information flag. This word has the packed format g. This word is optional. If this word is not entered, g = 0 is used.
- The digit g specifies whether noncondensable gas is present for this system. g = 0 specifies that noncondensable gas is present for this system. g = 1 specifies that noncondensable gas is not present for this system. If g = 1 (no noncondensable) in a system and if the digit t = 4, 5, or 6 in the hydrodynamic volume component control word gbt, an input error will result.
- If the 99 option in CARD 1 is on, g = 2 specifies that noncondensable gas is present and the iterative scheme is used. g = 3 specifies that noncondensable gas is not present and the iterative scheme is used.
- W6-9(A) Thermodynamic property file name of the system. This optional alphanumeric entry can be used to enter the file name of the thermodynamic property file of the system. Up to thirty-two characters may be entered as one alphanumeric field. (The code internally treats the field as up to four eight-character words.) The default file name for the thermodynamic property file is tpfh2o for H2O, tpfd2o for D2O, tpfh2onew for H2ONEW, tpfhe for helium, tpfco2 for CO₂, etc.

2.16 Self-Initialization Option Control

These cards are optional, are not needed, and are only used as a cross-check on the controllers specified in Section 14. Data supplied on these cards are used to invoke the self-initialization option. These data describe which and how many of each controller will be used. To retain generality and flexibility, the self-initialization option does not require that the steady-state and nearly-implicit solution scheme options be concurrently turned on. However, this is the recommended procedure. These latter options are invoked

through input data cards 100 and 201 through 299. In addition to the data cards described below, the user must furnish data on the controllers to be used, as described in Section 14.

2.16.1 Self-Initialization Control, 140

This card specifies the number and type of controllers desired.

W1(I) Number of pump controllers.

W2(I) Number of steam flow controllers.

W3(I) Number of feedwater controllers.

or

W1(A) DELETE or DISCARD. The command DELETE is allowed only in RESTART problems, and self-initialization control must be existing at the time of restart. The DELETE command deletes all controllers.

2.16.2 Self-Initialization Pump Controller and Identification, 141-142

These cards establish the relationship between the pump number and the number of the pump controller. For each pump so referenced, the user *must* use the time-dependent pump velocity option. For pump component card CCC6100, Words 2 and 3 must be the alphanumeric and numeric parts for the pump controller. The time-dependent pump velocity data (pump component cards CCC6100 through CCC6199) should be input so that the search variable and pump velocity are related by a straight line through the origin with a slope of 1.

W1(I) Component number of pump number.

W2(I) Controller identification number for pump number 1.

W3(I) Component number of pump number 2.

W4(I) Controller identification number for pump number 2.

A maximum of six pump/controller pairs may be entered.

2.16.3 Self-Initialization Steam Flow Controller Identification, 143-144

These cards establish the relationship between the steam flow control valve number and the steam flow controller number.

W1(I) Component number of steam flow control valve number 1.

- W2(I) Controller number of steam flow controller for steam flow control valve number 1.
- W3(I) Component number of steam flow control valve number 2.
- W4(I) Controller number of steam flow controller for steam flow control valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above the valve component is assumed to be the control component. However, the user is not constrained to use a valve and may use a pump or a time-dependent junction. CAUTION: only a servo valve, a time-dependent junction, or a pump may be used, or a diagnostic error will result.

2.16.4 Self-Initialization Feedwater Controller Identification, 145-146

These cards establish the relationship between the feedwater valve number and the feedwater controller number.

- W1(I) Component number of feedwater valve number 1.
- W2(I) Controller id number of the feedwater controller for feedwater valve number 1.
- W3(I) Component number of feedwater valve number 2.
- W4(I) Controller id number of the feedwater controller for feedwater valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above it is assumed that a valve component is the control component. However, the user is not constrained to use a valve and may use a pump or time-dependent junction. CAUTION: only a servo valve, time-dependent junction, or a pump is allowed, or a diagnostic will result, such as a time-dependent junction with the controller output used as the independent variable in place of time.

2.16.5 Pressure and Volume Control Component Identification, 147

This card identifies the component number, connection data, and pressure level for the time-dependent volume that is to provide pressure and volume control during the self-initialization null transient.

- W1(I) Component number of time-dependent volume that replaces the pressurizer.
- W2(I) Component number to which the above time-dependent volume is connected; CAUTION: only a single-junction is allowed or an error will result.
- W3(R) Desired steady-state pressure.